



SOUTHEAST UNIVERSITY
Meeting the Challenges of Time

Department Of CSE

CSE Assignment

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Course name : CSE LAB

Course code : CSE362

Section : 6

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Source code:

```
#include <stdio.h>

#include <limits.h>

#include <stdbool.h>

#define AGING_THRESHOLD 5

struct Process {
    int pid;
    int arrival_time;
    int burst_time;
    int priority;

    int remaining_time;
    int completion_time;
    int turnaround_time;
    int waiting_time;
    int response_time;

    bool started;
    int current_wait;
};

int main() {
    int n;
```

```
printf("Enter number of processes: ");
```

```
scanf("%d", &n);
```

```
struct Process p[n];
```

```
for (int i = 0; i < n; i++) {
```

```
    p[i].pid = i + 1;
```

```
    printf("Enter arrival time, burst time, and priority for P%d: ", i + 1);
```

```
    scanf("%d %d %d",
```

```
        &p[i].arrival_time,
```

```
        &p[i].burst_time,
```

```
        &p[i].priority);
```

```
    p[i].remaining_time = p[i].burst_time;
```

```
    p[i].started = false;
```

```
    p[i].current_wait = 0;
```

```
}
```

```
int current_time = 0;
```

```
int completed = 0;
```

```
float total_waiting_time = 0;
```

```
float total_turnaround_time = 0;
```

```
float total_response_time = 0;
```

```
printf("\n--- Starting Execution (Aging Threshold: %d ticks) ---\n",
AGING_THRESHOLD);
```

```
while (completed < n) {
```

```
    int selected = -1;
```

```
    int highest_priority = INT_MAX;
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (p[i].arrival_time <= current_time && p[i].remaining_time > 0) {
```

```
            if (p[i].priority < highest_priority) {
```

```
                highest_priority = p[i].priority;
```

```
                selected = i;
```

```
            }
```

```
        else if (p[i].priority == highest_priority) {
```

```
            if (p[i].arrival_time < p[selected].arrival_time) {
```

```
                selected = i;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
if (selected == -1) {
```

```
    current_time++;
```

```
} else {
```

```
if (p[selected].started == false) {  
    p[selected].response_time =  
        current_time - p[selected].arrival_time;  
  
    p[selected].started = true;  
}  
  
p[selected].remaining_time--;  
current_time++;  
  
if (p[selected].remaining_time == 0) {  
    p[selected].completion_time = current_time;  
  
    p[selected].turnaround_time = p[selected].completion_time -  
p[selected].arrival_time;  
  
    p[selected].waiting_time = p[selected].turnaround_time -  
p[selected].burst_time;  
  
    total_waiting_time += p[selected].waiting_time;  
    total_turnaround_time += p[selected].turnaround_time;  
    total_response_time += p[selected].response_time;  
  
    completed++;  
}  
}
```

```

for (int i = 0; i < n; i++) {
    if (i != selected && p[i].arrival_time <= (current_time - 1) && p[i].remaining_time
> 0) {
        p[i].current_wait++;

        if (p[i].current_wait >= AGING_THRESHOLD) {
            p[i].priority--;
            p[i].current_wait = 0;
            printf("[Time %d] AGING: Process P%d priority improved to %d due to
starvation.\n", current_time, p[i].pid, p[i].priority);
        }
    } else if (i == selected) {
        p[i].current_wait = 0;
    }
}
}

```

```

printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\tRT\n");

```

```

for (int i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid,
        p[i].arrival_time,
        p[i].burst_time,
        p[i].priority,

```

```

        p[i].completion_time,
        p[i].turnaround_time,
        p[i].waiting_time,
        p[i].response_time);
    }

    printf("\nAverage Waiting Time = %.2f", total_waiting_time / n);
    printf("\nAverage Turnaround Time = %.2f", total_turnaround_time / n);
    printf("\nAverage Response Time = %.2f\n", total_response_time / n);

    return 0;
}

```

Output of Priority :

```

nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ vim priority.c
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ gcc prioieity.c -o priority
cc1: fatal error: prioieity.c: No such file or directory
compilation terminated.
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ gcc priority.c -o priority
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ ./priority
Enter number of processes: 5
Enter arrival time, burst time, and priority for P1: 0 10 20
Enter arrival time, burst time, and priority for P2: 1 2 3
Enter arrival time, burst time, and priority for P3: 2 3 4
Enter arrival time, burst time, and priority for P4: 2 1 4
Enter arrival time, burst time, and priority for P5: 2 3 1

PID    AT    BT    PR    CT    TAT    WT    RT
P1      0    10    20    19    19     9     0
P2      1     2     3     6     5     3     0
P3      2     3     4     9     7     4     4
P4      2     1     4    10     8     7     7
P5      2     3     1     5     3     0     0

Average Waiting Time = 4.60
Average Turnaround Time = 8.40
Average Response Time = 2.20
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ |

```

Output of Aging:

```
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ vim aging.c
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ gcc aging.c -o aging
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ ./aging
Enter number of processes: 5
Enter arrival time, burst time, and priority for P1: 0 10 20
Enter arrival time, burst time, and priority for P2: 1 2 3
Enter arrival time, burst time, and priority for P3: 2 3 4
Enter arrival time, burst time, and priority for P4: 2 1 4
Enter arrival time, burst time, and priority for P5: 2 3 1

--- Starting Execution (Aging Threshold: 5 ticks) ---
[Time 6] AGING: Process P1 priority improved to 19 due to starvation.
[Time 7] AGING: Process P4 priority improved to 3 due to starvation.

PID    AT    BT    PR    CT    TAT    WT    RT
P1      0    10    19    19    19     9     0
P2      1     2     3     6     5     3     0
P3      2     3     4    10     8     5     4
P4      2     1     3     8     6     5     5
P5      2     3     1     5     3     0     0

Average Waiting Time = 4.40
Average Turnaround Time = 8.20
Average Response Time = 1.80
nazmul@DESKTOP-09D7DST:~/desktop/cse362.6/documents/priority$ |
```

Explanation:

- Every process has a current_wait counter. If a process is waiting in the queue, its counter goes up by 1 every tick.
- If that counter hits 5 (AGING_THRESHOLD), we subtract 1 from its priority number (which actually *increases* its priority power) and reset the counter back to zero. If the process gets the CPU, its wait counter resets to zero immediately.

```
if (i != selected && p[i].arrival_time <= (current_time - 1) &&
    p[i].remaining_time > 0)
```